WEEK 3 OF 15 · STAGE 1: FOUNDATIONS

Energy primer

Where emissions come from, in four numbers

New R skill dplyr: filter, select, mutate, group_by, summarise; the pipe

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	**Week 3** **Energy primer**	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Follow energy from primary source to the service people want
2. Write the Kaya identity and explain each of its four factors
3. Use ratios to see which factor drove a country's emissions
4. Use dplyr verbs and the pipe to filter and compute new columns

? **Big question** The CO2 from eight billion people's daily choices can be written as four numbers multiplied together. Which ones can policy change?

Four numbers behind all CO2

The CO2 from eight billion people's daily choices can be written as four numbers multiplied together. Two of them we hope will keep rising.



If population and income keep growing, what must happen to the other two?

AN EVERYDAY PICTURE

Energy is like water through leaky pipes



Water leaves a reservoir, passes a treatment plant, flows through pipes and comes out of your tap. Some is lost at every stage. Energy is the same: coal or sunlight is turned into electricity, sent through wires, and finally lights your desk.

What it means for models

1. You do not want energy, you want the service
2. Some energy is lost at each step
3. Two levers: cleaner supply, or less energy per service

HOW IT WORKS

From resource to service in four steps

1

Primary

Coal, oil, gas, uranium, sun, wind, biomass

2

Conversion

Power plants and refineries make usable forms

3

Final

Electricity, fuels and heat delivered to users

4

Services

Warm rooms, light, travel, steel, cement

Clean up the left side (supply), or need less energy for the same service (right side).

Worked example: losses along the chain

Problem Start with 100 units of coal. The power plant is 40% efficient and the grid loses 10%. How much reaches homes?

1. Power plant: 100 × 0.40 = 40 units of electricity
2. Grid: 40 × 0.90 = 36 units reach homes
3. Lost: 100 − 36 = 64 units, about two thirds
4. So 1 unit saved at home saves 100 ÷ 36 ≈ 2.8 units of coal

Answer

36 units reach homes; saving 1 unit at home saves about 2.8 units of coal

KEY POINTS

The Kaya identity

- $CO2 = P \times (G \div P) \times (E \div G) \times (C \div E)$
- P = population; G = GDP (the size of the economy)
- G ÷ P = income per person
- E ÷ G = energy intensity; C ÷ E = carbon intensity
- Each top cancels the bottom of the next, so it is always true

Worked example: a country called Avalon

Problem Avalon (invented) has 100 million people, $1,000 billion GDP, 10 EJ of energy and 600 Mt of CO2. What are its four factors?

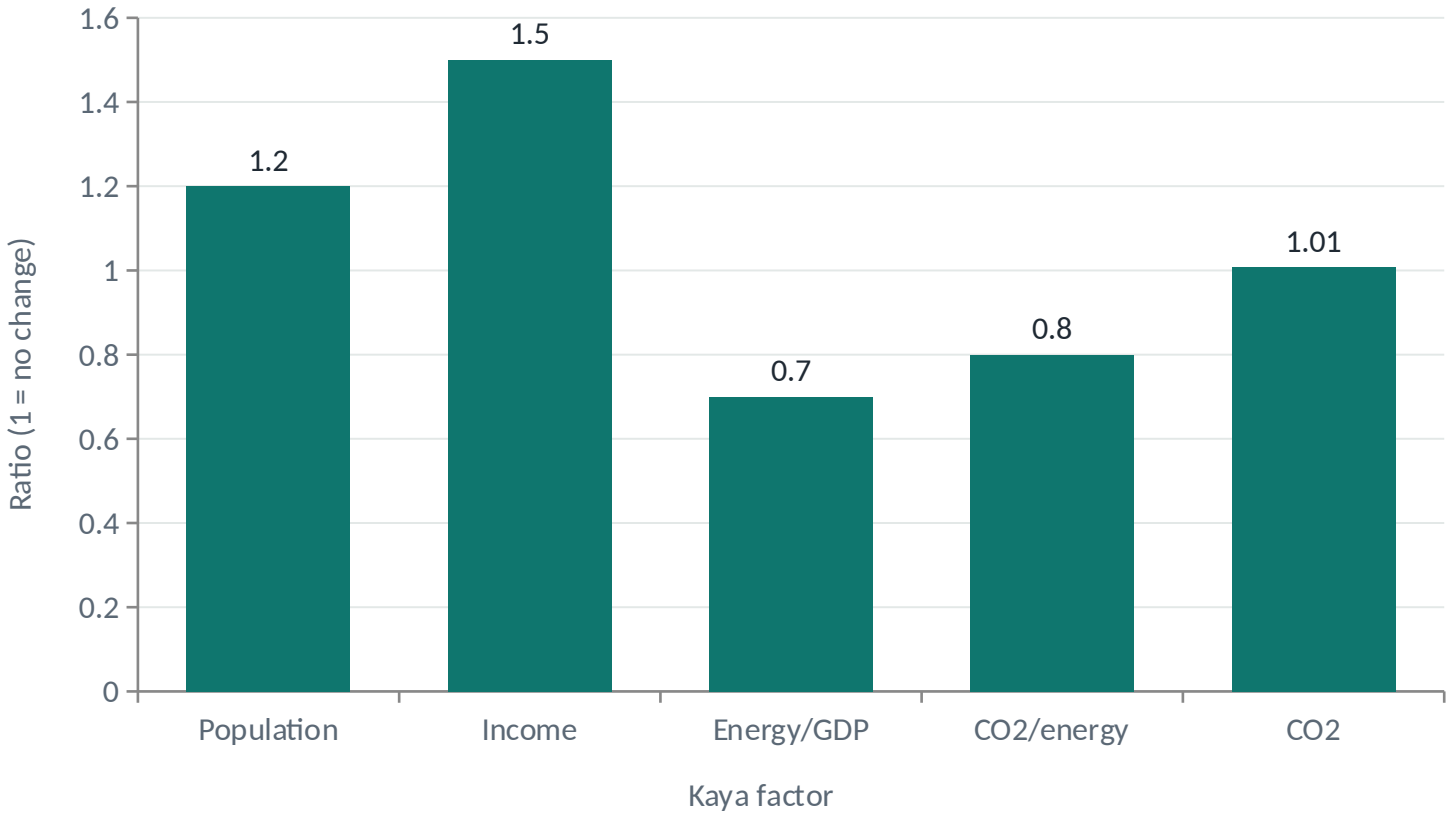
1. Income: $1,000 bn ÷ 100 m = $10,000 per person
2. Energy intensity: 10 EJ ÷ $1,000 bn = 10 MJ per dollar
3. Carbon intensity: 600 Mt ÷ 10 EJ = 60 kg CO2 per GJ
4. Check: 100 m × $10,000 × 10 MJ/$ × 0.06 kg/MJ = 600 Mt

Answer

Income $10,000; 10 MJ per $; 60 kg CO2 per GJ. The pieces rebuild 600 Mt.

PICTURE THE NUMBERS

Avalon 30 years later: what drove CO2?



Invented numbers. Lab 3 makes this chart for a real country.

Read it like this

1. Population ×1.2 and income ×1.5 pushed CO2 up
2. Energy intensity ×0.7 and carbon intensity ×0.8 pulled it down
3. 1.2 × 1.5 × 0.7 × 0.8 = 1.008: almost flat

Quick check

Over 20 years: population ×1.1, income ×2.0, energy intensity ×0.6, carbon intensity ×0.9. What happens to CO2?

- **A** Falls by about 10%
- **B** Stays the same
- **C** Rises by about 19%
- **D** Doubles

Quick check: answer

Over 20 years: population ×1.1, income ×2.0, energy intensity ×0.6, carbon intensity ×0.9. What happens to CO2?

A. Falls by about 10%

B. Stays the same

C. **Rises by about 19%** ✓

D. Doubles

WATCH OUT

Common mix-ups

Not quite	Better
Add the percentage changes: +20%, +50%, −30%, −20% gives +20%.	Kaya factors multiply: 1.2 × 1.5 × 0.7 × 0.8 = 1.008, only +0.8%.
Energy intensity and carbon intensity are the same thing.	Energy intensity is energy per dollar (efficiency). Carbon intensity is CO2 per unit of energy (clean supply).
Kaya explains why emissions changed.	It describes how they changed. The causes behind each factor need a model.

The pipe: read it as 'and then'

- filter keeps rows; select keeps columns
- mutate makes new columns from old ones
- == compares; a single = assigns

```
mine <- kaya |>
  filter(country == "Japan") |>
  select(year, pop, gdp, energy, co2) |>
  mutate(
    income   = gdp / pop,
    e_intens = energy / gdp,
    c_intens = co2 / energy
  )
```

NEW WORDS THIS WEEK

Words to add to the class glossary

Primary energy Energy as found in nature before conversion: coal, oil, gas, uranium, sunlight, wind, biomass.

Final energy Energy delivered to users after conversion: electricity, petrol, heat.

Energy intensity Energy used per dollar of GDP. Falls when the economy becomes more efficient.

Carbon intensity CO2 released per unit of energy. Falls when energy comes from cleaner sources.

Kaya identity CO2 = population × income per person × energy intensity × carbon intensity. Always true by arithmetic.

Lab 3: Kaya decomposition with dplyr

Break one country's emissions into population, income, energy intensity and carbon intensity, and find which factor drove the change.

Script: labs/lab3.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

LAB · STEP 1 OF 4

Pick a country and compute factors

Replace Japan with your own country. Run after each line so you can see what changed.

```
kaya <- read_csv("data/kaya_data.csv")
mine <- kaya |>
  filter(country == "Japan") |>
  select(year, pop, gdp, energy, co2) |>
  mutate(
    income   = gdp / pop,
    e_intens = energy / gdp,
    c_intens = co2 / energy
  )
```

Compare the last year with the first

summarise() turns many rows into one. last() ÷ first() gives the ratio for each factor.

```
change <- mine |>
  arrange(year) |>
  summarise(
    pop      = last(pop) / first(pop),
    income   = last(income) / first(income),
    e_intens = last(e_intens) / first(e_intens),
    c_intens = last(c_intens) / first(c_intens),
    co2      = last(co2) / first(co2)
  )
change
```

Check the identity

The four ratios must multiply to the CO2 ratio. If they do not, there is a typo somewhere.

```
change$pop * change$income *
  change$e_intens * change$c_intens
change$co2
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Three countries at once

%in% keeps several countries. group_by(country) makes summarise work once per country.

```
kaya |>
  filter(country %in% c("Japan", "India", "Brazil")) |>
  mutate(income = gdp / pop,
         e_intens = energy / gdp,
         c_intens = co2 / energy) |>
  arrange(year) |>
  group_by(country) |>
  summarise(co2 = last(co2) / first(co2),
            income = last(income) / first(income))
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

LAB · CHECK YOUR WORK

Did it work?

You should see

- Your four factor ratios multiply to the CO2 ratio
- You can name the factor that grew most in your country
- The three-country table has one row per country

Worksheet questions

1. Which Kaya factor rose most in your country, and which fell most?
2. Explain in two sentences why adding the percentage changes gives the wrong answer.
3. Compare your country with a neighbour's: which lever (efficiency or clean energy) did each rely on?

RECAP

Take-home messages

1. Energy flows from primary sources through conversion to the services people want, with losses
2. Kaya: CO2 = population × income × energy intensity × carbon intensity
3. The factors multiply, so compare ratios, not added percentages
4. Policy mainly works on the last two: efficiency and clean energy

Next: week 4

Week 4 adds the economics: how prices balance markets, what a carbon price does, and why the future gets discounted.

Practise

Interactive course, week 3: try the kaya explorer and the 5-question quiz.

Readings: IPCC AR6 Working Group III (2022), Summary for Policymakers: section on emissions trends and drivers; R for Data Science: data transformation chapter